

Section 3: Networking Basics

Task 3.1: Network Concepts

1. OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual framework used to understand how network communication works.

7 Layers of OSI Model

1. **Physical Layer**
Responsible for transmission of raw bits. Example: Ethernet cables.
2. **Data Link Layer**
Handles MAC addressing and error detection. Example: Ethernet protocol.
3. **Network Layer**
Responsible for routing and IP addressing. Example: IP.
4. **Transport Layer**
Provides reliable data transfer. Example: TCP, UDP.
5. **Session Layer**
Manages sessions between applications.
6. **Presentation Layer**
Data formatting, encryption, compression.
7. **Application Layer**
User interface for network services. Example: HTTP, FTP, DNS.

Note: TCP operates at the Transport Layer (Layer 4).

2. IP Addressing

Difference between Public and Private IP

Public IP	Private IP
Used on the internet	Used inside local networks
Globally unique	Can be reused
Assigned by ISP	Assigned by router

Private IP Ranges

- Class A: 10.0.0.0 – 10.255.255.255
- Class B: 172.16.0.0 – 172.31.255.255
- Class C: 192.168.0.0 – 192.168.255.255

192.168.1.100 is a Private IP address.

3. Protocols

Port Mapping

Port	Service
22	SSH
80	HTTP
443	HTTPS
3306	MySQL

TCP vs UDP

TCP	UDP
Connection-oriented	Connectionless
Reliable	Faster but unreliable
Error checking present	No guarantee of delivery

NAT (Network Address Translation)

NAT is a method used to convert private IP addresses into a public IP address so devices can access the internet.

Why NAT is needed:

- Conserves IP addresses
- Adds security by hiding internal network

Task 3.2: Practical Network Analysis

Commands Used

```
ifconfig
route -n
netstat -tuln
ping google.com
traceroute google.com
```

Explanation of Results

1. IP Address

The IP address is assigned to the system by the router. Example: 192.168.x.x (Private IP).

2. Default Gateway

The default gateway is the router's IP address that connects the system to the internet. Example: 192.168.x.1.

3. DNS Server

DNS servers translate domain names into IP addresses. Example: 8.8.8.8 (Google DNS).

4. Active Connections

Using netstat, we can see open ports and active services running on the system.

5. Ping Result

Ping confirms connectivity to google.com and shows latency.

6. Traceroute Result

Traceroute shows the path taken to reach google.com and number of hops.

Typical hops: 8–15 depending on network.

Conclusion

This practical analysis helped in understanding network configuration, connectivity testing, routing behavior, and real-world networking concepts.